

Fourteen Caption Users

Unit 3 · Topic 3.2 · THE PROBLEM

Suppose a streaming platform has 5,000 active accounts and says that $p = 0.08$ of them regularly use captions. An SRS of 100 accounts, selected without replacement, contains 14 regular caption users, so $\hat{p} = 0.14$.

Jules: “Because $n = 100 > 30$, $\hat{p}$ is approximately normal. I get $P(\hat{p} \geq 0.14) = 0.0135$, which proves caption use increased.”

Nora: “Because $np = 8 < 10$, a normal model is not justified. Therefore $\hat{p}$ is biased and this sample tells us nothing.”

A binomial calculator gives $P(X \geq 14) = 0.0282$ for $X \sim \text{Bin}(100, 0.08)$. This closely approximates sampling from 5,000 accounts.

Caption-use sample

Source	Accounts	Users	Rate
Population	5,000	400	0.08
SRS	100	14	0.14

FIRST TAKE — one sentence before you work the parts

Does finding 14 caption users instead of the expected 8 prove that use increased? Give one reason.

- DOK 1** (a) Identify p and verify the sample proportion $\hat{p} = 14/100$.
- DOK 2** (b) Find the mean and standard deviation of the sampling distribution of $\hat{p}$. Check the 10 percent condition and both Large Counts values.
- DOK 3** (c) In 3–4 sentences, explain what each student gets right and wrong. Use the condition checks and the supplied probability 0.0282. State what the sample suggests about caption use and why a failed normal-shape check does not make the estimate biased.

Self-paced bonus problem. Work (a)–(c) from this sheet; turn it in whenever you finish.